Conference Program

Wednesday, 14 October 2026 – Mustapha Stambouli University of Mascara

The First National Conference on Artificial Intelligence: Research Innovation and Systems

(NC-AIRIS'2026)

FOR ONLINE SESSIONS:

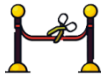
Meeting ID: , Passcode:

08:00-09:00 Welcome and Registration in Desk and Posters hang-up (Conference Hall)

Registration Desk

09:00 – 09:30 Opening Session

Room 1



Welcome Speech: *Dean of the Faculty*

Official Opening: *University Rector*

Welcome Speech and Program: *Conference & Program Chairs*

(Pr. Mohammed Kamal BENHAOUA, Dr. Hadya BOUFERA, Dr. Chérifa BOUDIA, Dr. Mohamed ZAGANE, Dr. Meriem MEDDEBER, Dr. Amine DAHANE)

Zoom Link: or Meeting ID: , Passcode:



09:30 – 10:00 Keynote Session 1 (Chairs: *Dr. Soraya SETTI AHMED* & *Dr. Ahmed El Amine DJELLOULI***)**

Strengthening Brain-Computer Interfaces Against Adversarial Threats: From Exposed Weaknesses to Robust AI Solutions.

Room 1

Pr. Chaker Abdelaziz KERRACHE, Vice-Rector of the Amar Telidji University of Laghouat.

Zoom Link: or Meeting ID: , Passcode:



10:00 – 10:30 Keynote Session 1 (Chairs: *Dr. Soraya SETTI AHMED* & *Dr. Ahmed El Amine DJELLOULI***)**

Artificial intelligence as a tool for research and innovation.

Room 1

Pr. Younes GUELLOUMA, Director of the AI Center at the Amar Telidji University of Laghouat.

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10:30 – 11:00 COFFEE BREAK & POSTER SESSION



Poster Session HALL
Chairs: *Dr. Faiza MAHI, Dr. Rachid KHELLADI, Dr. Abdellah KHELIL*
Zoom Link: or Meeting ID: , Passcode:



Oral Communications : Parallel Sessions

11:00-12:30 Session 1
Room 1
Chairs*: Dr. Soraya SETTI AHMED & Dr.* Meddeber Meriem
Zoom Link: or Meeting ID: , Passcode:

PaperID	Authors	Paper Title
40	*Bouchra Yamna Zekraoui, Fatima Debbat, Mohamed Amine Nemmich, Pablo Bermejo*	**LLM-Guided Initialization for Multi-Objective Bio-Inspired Feature Selection in Medical Diagnosis (On-site)**
80	*Kassouar Fatima, Abderrahmane Khiat*	**PNEUMO-AI: A Multimodal Deep Learning Framework Integrating Chest X-Rays and Clinical Symptoms for Pediatric Pneumonia Detection Using Unpaired Public Datasets (On-site)**
44	*Yamina Hachemi, Rabab Breik, Sarra Djerouane*	**Artificial Intelligence for Autism Spectrum Disorder Diagnosis (On-site)**
36	*Azzaz Amina; Boudia Cherifa*	**FedDerm-XAI: A Federated Explainable and Interoperable Multimodal Framework for Skin Lesion Triage in Multi-Center Teledermatology (On-site)**
59	*Djamal SEGHIER, Moustafa Maaskri, Mohamed Goismi*	**CLTD: Lightweight, Confidence-Gated ML-Assisted Threat Detection for Resource-Constrained NDN-IoT Networks (On-site)**
79	*Larbi keltoum, Kharroubi sahraoui*	**Beyond Risk Prediction: An Explainable Deep Learning Framework with Shap-Driven Recommendations for Diabetes Care (Online)**
21	*Aidoun Houcine, Barigou Fatiha, Atmani Baghdad*	**Comparative study to predict Mean Blood Pressure in Intensive Care using Deep Learning and Machine Learning Based on Time Series: LSTM and Ridge Regression models (Online)**
30	*Bedad Fatima, Nassima Bousahba, Lara Hibi, Hafsa Amina Mekadid*	**YOLO for Histopathology Classification: Architectural Rationale and Empirical Evidence from MSI/MSS Prediction (Online)**

11:00-12:30 Session 2
Room2
Chairs: *Dr. Mohammed* ZAGANE *& Dr. Mustapha SAHRAOOUI*
Zoom Link: or Meeting ID: , Passcode:

PaperID	Authors	PaperTitle
83	*Moustafa, MAASKRI; Djamal, Seghier; Mohamed, GOISMI; Mohamed, DEBBAB*	**Sentiment Analysis of Twitter Data: A Comparative Study of Traditional Vector Space Models and BERT-Based Approaches (On-site)**
54	*Kateb Hachemi Amar Amar, Benbrik Nihad, Bouazdia Rania, Hamdadou Djamel, Tahraoui Mohammed Amin*	**Multi-Objective Cluster-Head Selection in Wireless Sensor Networks Using Optimization by Morphological Filters (On-site)**
91	*Mahi Faiza, Cherifa Boudia, Kamel Benhaoua*	**AI-Enhanced Multi-Intersection Traffic Light Control Using GNN-DQN: A Comparative Study with Genetic and Grey Wolf Optimization (On-site)**
81	*Rachid Khalladi, Mohamed Zagane, Youcef Fekir, Abdallah Khelil, Abderrahime Mohammed Abdelouahab*	**Machine Learning-Based Detection of DDoS and Phishing Attacks for Cybersecurity Monitoring (On-site)**
78	*Djellouli Ahmed El Amine, Taieb Bouanni Chaimaa, Abdelhadi Setti*	**Formal Verification of an adaptive Negative Selection Algorithm for intrusion detection (On-site)**
35	*Mohammed Habib Diaf, Sihem Aissaoui, Mohammed Taher Amine Sahraoui*	**ConResvMixer-IDS: A Hybrid CNN-MLP-Mixer Architecture for Network Intrusion Detection (Online)**

11:00-12:30 Session 3
Room 3
Chairs*: Dr. Brahim Cherouati & Dr. Boudaieb Ahmed*
Zoom Link: or Meeting ID: , Passcode:

PaperID	Authors	PaperTitle
82	*Djamel Hamdadou, Ali Akrem Barka, Abdel Aziz Feraoun, Amar Kateb Hachemi Amar*	**An Intelligent OBD2/CAN-Based System for Vehicle Engine Fault Diagnosis and Real-Time Emission Monitoring (On-site)**
57	*Chérifa BOUDIA*	**From Passive Smartphone Use to Adaptive Learning: Designing and Evaluating an AI-Powered Mobile Tutor for Computer Programming Students (On-site)**
50	*Mohammed Ridha youbi, Mohammed Mesaadi*	**A Sovereign Multi-Agent RAG Architecture for Adaptive Education in Engineering (On-site)**

18	*Hadya Boufera, Ahlem Semmar, Marwa Rahal*	**Speak with Confidence: A Mobile Application for the Early Screening of Reading Dyslexia in Arabic-Speaking Children (On-site)**
26	*Mohammed Zagane, Rachid Khalladi, AbdAllah Khelil, Brahim Khaldi*	**Semantic Sink Discovery for Rule-Based SAST Using Code and Documentation Embeddings (On-site)**
93	*Meriem Zingla*	**Interpretable Association-Rule-Based Classification of Kidney-Related Health Status Using NHANES Questionnaire Data (On-site)**
90	*Reguieg, Seddik; Omari, Youcef; Mami, Mohamed Amine; Haffaf, Malik; Yousfi, Mostafa Jamal*	**Feature-Augmented Deep Reinforcement Learning for Sequential Pedagogical Role Assignment in Residency Training (On-site)**
71	*Seddik Boubakar, DEGHEM; LAHCENE, AID; kadda, mostefaoui*	**Minority-Aware Feature Selection for Compact Network Traffic Classification Using CatBoost (On-site)**

12:30 –14:00 LUNCH



Online Communications: 3 Parallel Sessions

14:00–15:30 Session 4
Room 1
Chairs : *Dr. Chokri MOKHTARI & Dr. Rachid RIMANI*
Zoom Link: or Meeting ID: , Passcode:

PaperID	Authors	PaperTitle
63	*Hassina Bouressace, Tarek Ferhi, Mohamed Reda Younsi, Amdjed Boumaza*	**DarLink: A Localized Web Platform for Vacation Rentals in Algeria with ID-Card OCR Registration, Local Card Payments, and Two-Sided Reviews (Online)**
77	*Zineb Deffas, Mounira Bouzahzah, Alaeddine Bellili, Hicham Bouabibsa*	**A Hierarchical Machine Learning Framework for Infant Cry Classification (Online)**
22	*Abdennour Redjaibia, Samia Drissi, Karima Boussaha, Ayemen Chibouti*	**Auditing False Discovery Rate-Controlled Fairness Disparities in Deep Learning-Based MOOC Completion Prediction (Online)**
96	*Mehdi Khitas, Moncef Khitas, Bouthaina Satour, Asma Hamari*	**Design and Implementation of a Real-Time Multimodal System for Algerian Sign Language Recognition and Translation (Online)**
29	*Aya Lamiri, Nadir Farhi, Meriem Bouali*	**Large Pretrained Models for Reward Design in Reinforcement Learning: A Comparative Study with Applications to Autonomous Driving (Online)**
13	*Saida Lemnadjlia*	**Intelligent Semantic Compression of Medical Images Using Deep Learning (Online)**

37	*Nour El Houda Bourai, Akila Djebbar, Hayet Farida Merouani*	**Does Compression Reach the Margins of Clinical AI? A Rapid Review of Under-Explored Imaging Domains (Online)**
85	*Yamina Bordjiba, Ali Seridi*	**Automatic Chest X-Ray Report Generation: Methods, Evaluation, Challenges, and Future Directions (Online)**
5	*Azizi Selma*	**Machine Learning-Driven Cybersecurity Framework For Secure Smart Manufacturing in Industrial IoT Environments (Online)**
11	*Nassima BOUSAHBA, Mahammed Abdelmadjid ALLALI, Asmaa KEFKEF, Sarra LAHMAR, Rachid BECHAR.*	**A Hybrid Machine Learning and Wavelet-Attention Framework for Multi-Horizon Agricultural Drought Forecasting in the Chlef Province, Algeria (Online)**

14:00–15:30 Session 5
Room 2
Chairs : *Dr.* Youcef FEKIR *&* Djamel HAMDADOU
Zoom Link: or Meeting ID: , Passcode:

PaperID	Authors	PaperTitle
1	*Belaidi Belaid, Krim Teyabe, Haimouri Abdelbasset, Bousaid Mohamed*	**Development of a Multifunction Smart Transfer Chair for Healthcare Assistance Based on ESP32 Embedded Control *(Online)***
70	*Ismahene Marouf, Asma Saighi, Sofia Kouah, Zakaria Laboudi*	**Quantifying Network Readiness for Healthcare Services: From 5G to 6G *(Online)***
24	*Ahlam BOUSSADIA, Youcef BOUGRIOU*	**An FDQ-CBR Protocol for IoVs *(Online)***
19	*Abdelbacet Brahmia, Brahim Farou, Rania Farah, Ala Eddine Benrazek, Zineddine Kouahla*	**A Parallel Enhanced TCB-Tree for Efficient and Scalable Data Retrieval in Smart Grids *(Online)***
84	*Ikrame Boubaker*	**A Comparative Study of AraBERT, XLM-RoBERTa, and DziriBERT on Algerian Sentiment Analysis *(Online)***
56	Imene Achouri, Youssouf Achouri, Malika Babes	**Elliptic Curve Cryptography for Secure IoT Communications: A Lightweight Performance Evaluation *(Online)***
76	*Abdelmalek Bakiri, Samir Fenanir, Lakhdar Goudjil*	**Train–Measure–Shrink: A Constraint-Aware Loop for Deploying Gradient-Boosted Intrusion Detectors on TinyML Devices *(Online)***
58	*Youcef Djeghoubbi, Ahmed Korichi, Hadya Boufera*	**PriRo-FedNIDS: A Privacy-Preserving and Byzantine-Resilient Federated Learning Framework for Network Intrusion Detection under Non-IID Traffic *(Online)***

20	*Djawhara Benchaira, Okba Tibermacine*	**A Comprehensive Review of Threat Models and Evasion Attacks in Production Machine Learning Applications *(Online)***
46	*Abdelhamid, BENHAMMOU*	**LSTM versus Transformer Architectures for Predictive Cloud Resource Allocation in Hospital Information Systems: A Comparative Analysis of Recent Approaches *(Online)***

14:00-15:30 Session 6
Room3
Chairs : *Dr.* Rachid KHALLADI *& Dr.* Abdallah *KHELIL*
Zoom Link: or Meeting ID: , Passcode:

PaperID	Authors	PaperTitle
38	*Sofiane Aouag, Adel Berradj, Hanane Merouani*	**AI-Driven Vehicle Safety and Emergency Decision Framework for Smart Cities (Online)**
43	*Zenati Yassine, Tlemsani Lotfi*	**Vision-Guided Selective Laser Weeding with YOLOv8 and a Linear Delta Robot: Perception, Kinematic Modeling, and PID Control (Online)**
28	*Amira Bourechak, Noureddine Seddari*	**Recent Advances in Artificial Intelligence for Traffic Forecasting: Emerging Trends and Open Challenges (Online)**
86	*Sarah Younes, Soumia Bourbia*	**Cross-Gated Spatio-Temporal Fusion via GCN and GRU for Traffic Flow Prediction (Online)**
27	*Ali Benyounes, Brahim Farou*	**Object-Level Targeted Few-Shot Learning for Imbalanced Wheat Grain Quality Classification (Online)**
4	*Meradi chaima, Aouag Sofiane, Beloucif Assia*	**A Survey of Security Verification Approaches at the Software Architecture Level: Techniques, Trends, and Open Challenges (Online)**
47	*Nour El Yakine Guerras, Afifa Ghenai, Samir Brahim Belhaouari*	**Toward Privacy-Preserving Distributed Intrusion Detection in IoMT: A Survey and Lessons Learned (Online)**
15	*Hanene Merouani, Sonia-Sabrina Bendib, Hamouma Moumen, Youssouf Achouri, Meriem Abidi*	**Hybrid Orchestration of AI, Blockchain, and Quantum Paradigms for Fog–IoT Systems (Online)**
45	*Ikram Maouche, Sadek Labib Terrissa, Mohamed Fouad Maouche, Noureddine Zerhouni*	**Explainable Cost-Sensitive CatBoost for Breast Cancer Recurrence Prediction (Online)**
25	*Nihel Fatima Baarir, Samir Bourekkache*	**Early Identification of Students at Risk (Online)**

16:30 – 17:00 Closing Ceremony & Acknowledgments, Trophies, Group Photo

Conference Program

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The First National Conference on Artificial Intelligence: Research Innovation and Systems (NC-AIRIS'2026)

POSTER SESSION: Chairs: Dr. Ahmed Amine DJELLOULI Dr. Hayat MEDDEBER & Dr. Faiza Mahi & Dr. Sabrine Abid

FOR ONLINE SESSIONS: Zoom Link:

ID	Authors	Affiliation	PaperTitle
88	Miloud Dahane, Amel Benlaldj, Seddik Reguig	University of Oran	Toward Automatic CO2 Monitoring to Demonstrate Its Fertilizing Effect
49	Abderrahmane Louni, Samir Ghouali, Mokhtar Besseghier, Abdelkader Horch	University of Mascara	Machine Learning for Thyroid Disease Prediction: Performance and Interpretability
61	Mawloud Titah, Ibrahim El Khalil Mekki	University of Oran 2 Mohamed Ben Ahmed, Oran, Algeria	Knowledge Graph Modeling and Semantic Rule Inference for Intelligent Hospital Inventory Governance and Logistics Interoperability
100	Bendouma Houaria, Bechar Hassane, Baba Hamed Amel	Abou Bekr Belkaid University of Tlemcen	Real-Time Adaptive Luminance Enhancement of Multimodal Medical Images Using FPGA-Based Rule Decision Logic
75	Ihsene Zaidi, Said Brahimi, Soufiane Boulehouache	University of Skikda 20 Août 1955, Skikda, Algeria	Triplet Selection for Robust Embedding Learning in Person Re-Identification
51	Islam Hadj Mohammed Smail, Chaker Abdelaziz Kerrache, Fatima Zohra Bousbaa	University of Laghouat	CBAM-ResCNN: A Residual CNN with Integrated CBAM for Blight Disease Detection in Solanum tuberosum Leaves
89	Mounira Bouzahzah, Nourhane Boughera, Radhia Cherroum	Abd Elhafid Boussouf University Center, Mila, Algeria	All Seasons: An Integrated IoT Framework for Smart Greenhouse Environmental Control and Early Apple Leaf Disease Detection
66	boukhezar, abderrahmane; Dahane, Amine; Bendouda, Djamila; Benameur, Rabaie	University of Oran 1 Ahmed Ben Bella, Algeria	A Trust-Aware Architecture for Wastewater Monitoring and Irrigation Decision Support
97	Abdallah Khelil	University of Mascara	A Hybrid Multi-Agent Intrusion Prevention Framework for Adaptive Network Threat Detection and Response
68	Mbarka Meriem Benikhlef, Amine Dahane, Rabaie Benameur, Sami Souihi	University of Oran 1 Ahmed Ben Bella, Algeria	A Survey on Secure and Trustworthy Respiratory Disease Detection: IoT, Federated Learning, and Blockchain
67	Lahmer Mohammed, Abdelghani Ghomari, Amine Dahane, Rabaie Benameur, Sami Souihi	University of Oran 1 Ahmed Ben Bella, Algeria	Accuracy–Latency Tradeoffs for Low-Light Object Detection on ExDark
101	Boudaieb Ahmed	University of Mascara	Covariance Matrices using Learning Vector Quantization: Comparison of Euclidean and Riemannian Approaches and Study of Congruence Invariance
73	Samia Gharbi, Ahmed Saci, Mohammed Hadjarai, Noureddine Karkachi	University of Oran 1 Ahmed Ben Bella, Algeria	Predictive Modeling of Rhizospheric Bacterial Antagonists Based on AI for Sustainable Biocontrol and Precision Crop Protection